

Representational Geometry in Neural Networks

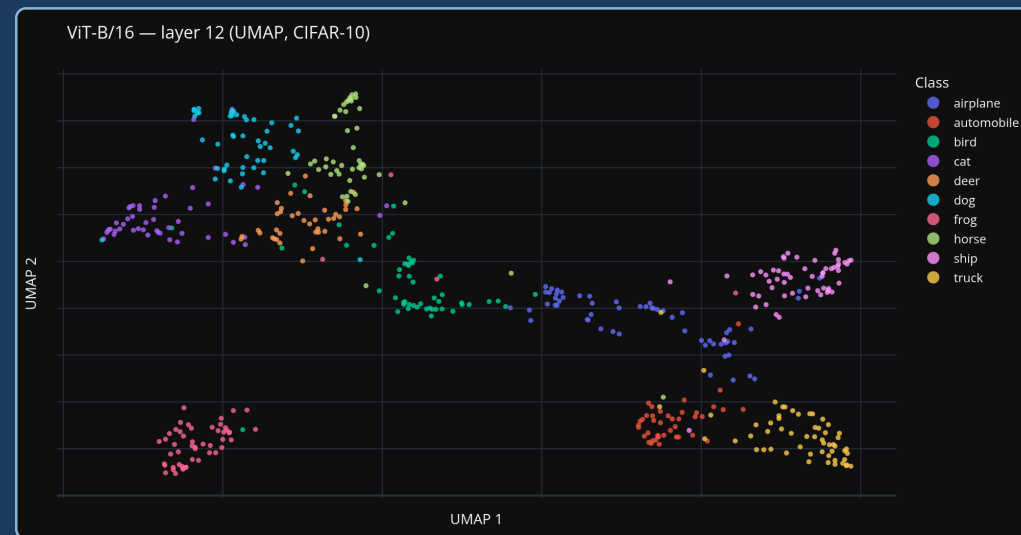
From Vision Transformers to RLHF-Aligned Language Models

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Summer Semester 2026

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ViT-B/16 · layer 12 · CIFAR-10 · real data

Part 1 — Cross-modal alignment

When a vision model and a language model train on the **same task**, does a shared representational structure emerge across modalities?

- **Hypothesis A:** cross-modal signal is new information → improves model
- **Hypothesis B:** models converge regardless — extra signal adds little

CKA before/after cross-modal fine-tuning · DINOv2 + Llama
· MS-COCO

Part 2 — RLHF geometry

Does RLHF geometrically transform a language model's embedding space — encoding human preference as a **linearly separable structure**?

- Does Instruct separate chosen/rejected responses that base cannot?
- How does separation evolve **layer by layer**?

UMAP + LinearSVC per layer · Llama-3-8B base vs. Instruct · Anthropic/hh-rlhf

Setup & Expected Findings

Part 1 — Cross-modal · DINOv2 + Llama · MS-COCO · multi-class prediction

Both models trained on **same task** (object category prediction) · measure CKA · fine-tune with loss term encouraging similarity to the other model's representation · compare

CKA high before fine-tuning: models converge inherently (**Platonic Representation Hypothesis**)

CKA rises after fine-tuning: cross-modal signal is genuinely new information

Layer-wise CKA map shows **where** shared meaning lives · UMAP + dashboard to visualize

Part 2 — RLHF geometry · Llama-3-8B base vs. Instruct

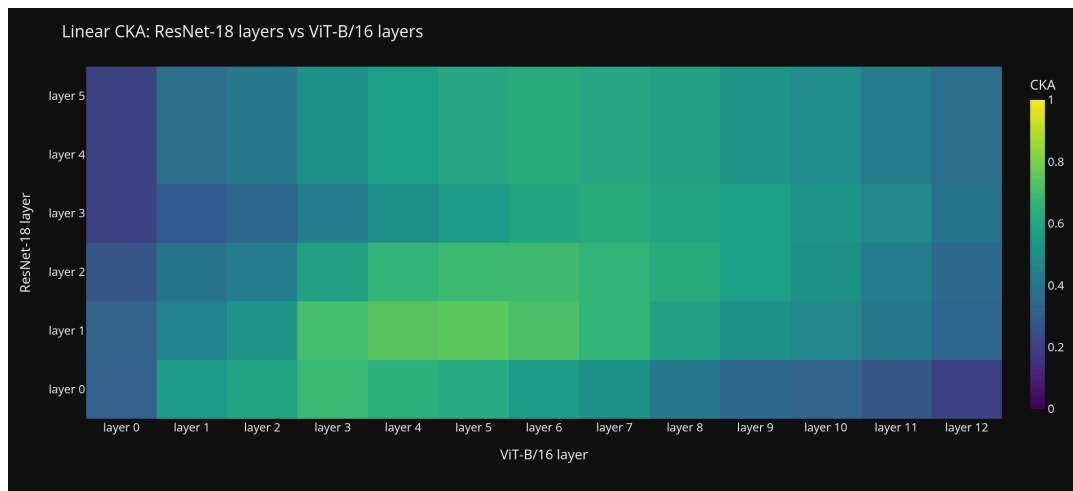
UMAP + LinearSVC per layer · Anthropic/hh-rlhf

Phase transition: base near-chance SVC — Instruct shows sharp jump, pinpointing **where** alignment lives

UMAP split: chosen/rejected mixed in base, cleanly separated in Instruct

Perspective: extend to RLHF training **snapshots** — plot cross-modal CKA vs. training step to see if alignment shifts **during** RLHF

He-Trott-Khosla 2025 · pile of LMs + vision models



Interactive Plotly Dash · CKA heatmap · UMAP layer slider · LinearSVC score

Next steps — toward July

- Cross-modal CKA: DINOv2 vs. Llama on MS-COCO pairs
- **CLIP (ViT-B/32)** as upper bound — explicitly trained cross-modal alignment
- Compare base vs. Instruct CKA scores

Project deliverable

The dashboard **is** the deliverable — interactive visualization of all findings: layer slider, CKA heatmaps, UMAP scatter, SVC score

He, Trott, Khosla (2025)

Shared Latent Representations across Vision and Language

arXiv:2509.20751 · **Anchor paper**

Kornblith et al. (2019)

Similarity of Neural Network Representations Revisited

ICML · arXiv:1905.00414 · **CKA method**

Christiano et al. (2017)

Deep RL from Human Preferences

NeurIPS · arXiv:1706.03741 · **RLHF foundations**

Kucukahmetler et al. (2026)

Relative Geometry of Neural Forecasters

TMLR · arXiv:2602.15676

Ouyang et al. (2022)

Training LMs to Follow Instructions with Human Feedback

NeurIPS · arXiv:2203.02155 · **InstructGPT / RLHF**

McInnes et al. (2018)

UMAP: Uniform Manifold Approximation and Projection

arXiv:1802.03426 · **Dim. reduction method**